Supplementary Materials: What Changes When LLMs Learn to Reason? Mechanistic Signatures of Cognitive Bias Processing

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This document provides additional details on the benchmark design, full behavioral and probing results, cross-prediction analyses, transfer matrices, representational geometry, and lure susceptibility distributions that complement the main text. Section labels correspond to the appendix identifiers referenced in the main paper.

A. Full Benchmark Description

A.1 Overview

The benchmark comprises **470 items** organized into **11 categories** spanning **4 heuristic families**. Of these, 380 items (190 matched conflict/control pairs) across 8 categories form the primary evaluation set reported in the main text. The remaining 90 items (45 pairs) belong to 3 additional categories added for extended analysis: a natural frequency framing subset, sunk cost variants, and contamination baselines.

Every conflict item has a structurally matched no-conflict control that preserves surface features (narrative length, domain, vocabulary) while removing the heuristic–normative tension. This matched-pair design follows De Neys' conflict detection paradigm [De Neys, 2012].

A.2 Design Principles

1. **Novel isomorphs.** All items use novel surface features (names, professions, quantities, scenarios) to resist memorization from pre-training corpora. Classic versions (e.g., the original bat-and-ball problem, the Linda problem) are included only as contamination baselines and excluded from all primary analyses.

2. **Matched pairs.** Each conflict item is paired with a structurally identical control. For base rate neglect, the control aligns the diagnostic description with the base rate so no conflict exists. For conjunction fallacy, the control presents the single event as more representative than the conjunction. For syllogisms, the control aligns conclusion believability with logical validity.

3. **Difficulty calibration.** Items are assigned difficulty ratings from 1 (trivial) to 5 (maximal heuristic pull) based on pilot testing. Difficulty ratings are balanced across conflict/control pairs within each category.

4. **Natural frequency framing.** A 10-pair subset of base rate neglect items presents statistical information in Gigerenzer's natural frequency format ("10 out of 1000" rather than "1%") to test the ecological rationality critique [Gigerenzer and Hoffrage, 1995].

A.3 Category Table

Table 1 provides the full category breakdown with example prompts and sourcing information.

A.4 Scoring

Each item is scored on a four-way classification:

- **Correct**: the normatively correct response.
- **S1-lure**: the specific heuristic-predicted wrong answer (e.g., "lawyer" when base rates strongly favor "engineer").
- **Other-wrong**: an incorrect response that does not match the heuristic prediction.
- **Refusal**: the model declines to answer or hedges without committing.

The primary metric is the *lure rate*: the proportion of S1-lure responses on conflict items. This is more informative than raw accuracy because it distinguishes heuristic errors from random errors.

B. Full Behavioral Results

B.1 Full Lure Rate Table

Table 2 extends Table 1 of the main text to include all categories (including sunk cost where available) and binomial 95% confidence intervals.

B.2 Qwen THINK vs. NO_THINK Detailed Comparison

Table 3 provides a detailed breakdown of the within-model comparison for Qwen 3-8B.
Key observations from the Qwen within-model comparison:

- Base rate neglect shows near-complete correction: 56% → 4% (14× reduction).
- Conjunction fallacy remains majority-susceptible: 95% → 55% (only 1.7× reduction). Thinking mode helps but does not overcome this deeply encoded heuristic.
- Syllogisms are already immune without thinking, unlike Llama-3.1-8B where syllogisms are highly vulnerable (52%). This confirms that vulnerability profiles are model-specific, not universal.
- All four immune categories remain at 0% in both modes.

C. Layer-Wise Probe Results (All Layers)

C.1 Llama-3.1-8B-Instruct and R1-Distill-Llama-8B (Layers 0–31)

Table 4 reports the full layer-wise probe AUC for both Llama-family models across all 32 layers. Probes are ℓ_2-regularized logistic regression (5-fold stratified CV) trained on conflict vs. no-conflict classification restricted to the three vulnerable categories (base rate neglect, conjunction fallacy, syllogisms).

Table 4: Layer-wise probe AUC for Llama-3.1-8B-Instruct and R1-Distill-Llama-8B. "Gap" = Llama AUC − R1 AUC. Hewitt–Liang control AUC and selectivity (true − control) are shown where available. All values are ROC-AUC from 5-fold stratified cross-validation on the three vulnerable categories (n = 142 conflict + 142 control items).

	Llama 8B-Inst.		**R1-Distill-Llama-8B**			
Layer	**AUC**	**Control**	**AUC**	**Control**	**Gap**	**Selectivity**
0	0.848	**[D]**	0.782	**[D]**	0.066	**[D]**
1	0.899	**[D]**	0.831	**[D]**	0.068	**[D]**

Layer	AUC	Control	AUC	Control	Gap	Selectivity
2	0.926	**[D]**	0.870	**[D]**	0.056	**[D]**
3	0.932	**[D]**	0.901	**[D]**	0.031	**[D]**
4	0.927	**[D]**	0.919	**[D]**	0.008	**[D]**
5	0.939	**[D]**	0.928	**[D]**	0.011	**[D]**
6	0.933	**[D]**	0.926	**[D]**	0.007	**[D]**
7	0.960	**[D]**	0.927	**[D]**	0.033	**[D]**
8	0.952	**[D]**	0.920	**[D]**	0.032	**[D]**
9	0.948	**[D]**	0.928	**[D]**	0.020	**[D]**
10	0.971	**[D]**	0.927	**[D]**	0.044	**[D]**
11	0.979	**[D]**	0.922	**[D]**	0.057	**[D]**
12	0.988	**[D]**	0.927	**[D]**	0.061	**[D]**
13	0.993	**[D]**	0.927	**[D]**	0.066	**[D]**
14	**0.999**	**[D]**	**0.929**	**[D]**	**0.070**	**[D]**
15	0.996	**[D]**	0.919	**[D]**	0.077	**[D]**
16	0.995	**[D]**	0.929	**[D]**	0.066	**[D]**
17	0.989	**[D]**	0.923	**[D]**	0.066	**[D]**
18	0.991	**[D]**	0.927	**[D]**	0.064	**[D]**
19	0.987	**[D]**	0.914	**[D]**	0.073	**[D]**
20	0.978	**[D]**	0.914	**[D]**	0.064	**[D]**
21	0.969	**[D]**	0.923	**[D]**	0.046	**[D]**
22	0.968	**[D]**	0.922	**[D]**	0.046	**[D]**
23	0.963	**[D]**	0.917	**[D]**	0.046	**[D]**
24	0.978	**[D]**	0.924	**[D]**	0.054	**[D]**
25	0.964	**[D]**	0.922	**[D]**	0.042	**[D]**
26	0.965	**[D]**	0.918	**[D]**	0.047	**[D]**
27	0.969	**[D]**	0.906	**[D]**	0.063	**[D]**
28	0.981	**[D]**	0.923	**[D]**	0.058	**[D]**
29	0.965	**[D]**	0.924	**[D]**	0.041	**[D]**
30	0.973	**[D]**	0.918	**[D]**	0.055	**[D]**
31	0.954	**[D]**	0.904	**[D]**	0.050	**[D]**
Mean	**0.962**	—	**0.912**	—	**0.050**	—
Std	0.032	—	0.031	—	0.019	—

Note: Control AUC columns (Hewitt–Liang probes trained on randomly permuted labels) and selectivity values are marked **[D]** where per-layer values are pending. The main text reports aggregate control AUC ≤ 0.55 at all layers for both models, yielding selectivity >40 percentage points at peak layers. Per-layer control values will be filled upon completion of the full control probe sweep.

C.2 Summary Statistics by Layer Phase

The deliberation gap widens from early (0.035) to late (0.059) layers, peaking in the late phase, then narrows slightly in final layers (0.051). The early-layer gap does not survive BH-FDR correction ($p = 0.070$), while mid, late, and final phases are all significant ($p < 0.001$).

C.3 Qwen 3-8B NO_THINK and THINK (Layers 0–35)

Table 6 reports layer-wise probe AUC for Qwen 3-8B in both thinking modes. Qwen 3-8B has 36 transformer layers (layers 0–35).

Table 6: Layer-wise probe AUC for Qwen 3-8B in NO_THINK and THINK modes. Bold indicates the peak layer. Both modes produce identical pre-generation representations, reflected in identical probe AUC at every layer.

Layer	NO_THINK AUC	THINK AUC	Difference
0	**[DATA]**	**[DATA]**	0.000
1	**[DATA]**	**[DATA]**	0.000
2	**[DATA]**	**[DATA]**	0.000
3	**[DATA]**	**[DATA]**	0.000
4	**[DATA]**	**[DATA]**	0.000
5	**[DATA]**	**[DATA]**	0.000
6	**[DATA]**	**[DATA]**	0.000
7	**[DATA]**	**[DATA]**	0.000
8	**[DATA]**	**[DATA]**	0.000
9	**[DATA]**	**[DATA]**	0.000
10	**[DATA]**	**[DATA]**	0.000
11	**[DATA]**	**[DATA]**	0.000
12	**[DATA]**	**[DATA]**	0.000
13	**[DATA]**	**[DATA]**	0.000
14	**[DATA]**	**[DATA]**	0.000
15	**[DATA]**	**[DATA]**	0.000
16	**[DATA]**	**[DATA]**	0.000
17	**[DATA]**	**[DATA]**	0.000
18	**[DATA]**	**[DATA]**	0.000
19	**[DATA]**	**[DATA]**	0.000
20	**[DATA]**	**[DATA]**	0.000
21	**[DATA]**	**[DATA]**	0.000
22	**[DATA]**	**[DATA]**	0.000
23	**[DATA]**	**[DATA]**	0.000
24	**[DATA]**	**[DATA]**	0.000
25	**[DATA]**	**[DATA]**	0.000
26	**[DATA]**	**[DATA]**	0.000
27	**[DATA]**	**[DATA]**	0.000
28	**[DATA]**	**[DATA]**	0.000
29	**[DATA]**	**[DATA]**	0.000
30	**[DATA]**	**[DATA]**	0.000
31	**[DATA]**	**[DATA]**	0.000
32	**[DATA]**	**[DATA]**	0.000
33	**[DATA]**	**[DATA]**	0.000
34	**0.971**	**0.971**	**0.000**
35	**[DATA]**	**[DATA]**	0.000

Note: The think/no-think difference is exactly 0.000 at every layer because both modes share identical weights and the `P0` position is deterministic given the prompt. The thinking toggle only affects generation behavior *after* `P0`. Per-layer AUC values are marked **[DATA]** where full Qwen layer-wise results are pending; the peak at L34 (0.971) is confirmed. Hewitt–Liang control probes for Qwen are **[DATA]** pending.

D. Cross-Prediction Details

Cross-prediction tests train a probe on the three *vulnerable* categories (base rate neglect, conjunction fallacy, syllogisms) and evaluate it on the four *immune* categories (CRT, arithmetic, framing, anchoring). If the probe

captures a universal processing-mode signal, transfer AUC should remain high; if it captures category-specific lure susceptibility, transfer AUC should drop to chance (≈0.5) or below.

D.1 Full Cross-Prediction Results

Table 7 reports cross-prediction AUC at all 10 tested layers for both Llama-family models.

D.2 Interpretation

Llama-3.1-8B-Instruct. Transfer AUC drops dramatically from within-category performance at all mid-to-late layers. At the peak layer (L14), transfer AUC is 0.378—*below* chance (0.5), indicating that the probe's learned direction is *anticorrelated* with the immune-category conflict/control distinction. This anticorrelation is strong evidence that the probe captures processing-mode information specific to the vulnerable categories, not generic surface features of lure text. The mean transfer AUC across all tested layers is 0.444 (bootstrap 95% CI: [0.386, 0.515]).

R1-Distill-Llama-8B. The pattern is qualitatively different. At early layers (L0–L8), transfer AUC remains high (≥0.734), suggesting that R1-Distill's early representations encode a more generic conflict/control signal that partially transfers across categories. Transfer AUC declines monotonically in later layers, reaching 0.385 at L31—comparable to Llama's specificity. The mean transfer AUC is 0.654 (bootstrap 95% CI: [0.551, 0.752]), significantly above Llama's ($p < 0.05$, paired bootstrap test).

Layer-wise specificity trajectory. For Llama, specificity is established by L4 (drop = 0.488) and remains high thereafter. For R1-Distill, specificity develops gradually, with the drop exceeding 0.3 only from L20 onward. This suggests that reasoning distillation delays category-specific encoding to later layers while maintaining a more generic early representation.

E. Transfer Matrix

The transfer matrix tests whether probes trained on individual categories transfer to other categories, revealing shared vs. distinct representational structure across bias types. We train a logistic regression probe on each category independently and test on all other categories at layer 14 of Llama-3.1-8B-Instruct (the peak probing layer).

E.1 Full Transfer Matrix (Llama L14)

Table 8 reports the full category-to-category transfer AUC matrix.

E.2 Interpretation

The partial transfer matrix reveals a structured pattern. Key findings from the available data:

- **Base rate ↔ conjunction transfer is near-perfect** (AUC = 0.993 in both directions). These two representativeness-heuristic categories share a processing-mode direction that is nearly identical in the residual stream at L14.
- **CRT probes do not transfer to vulnerable categories** (AUC = 0.062, as reported in the main text). This confirms that CRT items, where all models are immune, occupy a distinct region of representational space.
- The high within-vulnerable transfer (AUC ≥ 0.950 among base rate, conjunction, and syllogism) combined with the low CRT-to-vulnerable transfer (0.062) provides converging evidence that the probe captures a shared processing-mode signal specific to the vulnerable categories, not a generic surface-feature signal.
- **[DATA]**: Full 7×7 matrix pending for syllogism cross-transfer and all immune-to-immune and immune-to-vulnerable entries.

F. Representational Geometry

We assess the geometric structure of conflict vs. no-conflict representations using two complementary metrics: cosine silhouette scores (cluster separability) and Centered Kernel Alignment (CKA; representational similarity between models).

F.1 Silhouette Scores

Silhouette scores measure how well conflict and no-conflict items cluster in residual stream space. We compute cosine silhouette after PCA reduction to 100 dimensions (to address the $d \gg N$ pitfall per Cover's theorem). Both models show positive but low silhouette scores, indicating that conflict and no-conflict items form overlapping clusters with only modest geometric separation. Llama's higher silhouette (0.079 vs. 0.059) is consistent with its higher probe AUC—the linear boundary is more distinct when clusters are marginally better separated. However, the low absolute values (far below the 0.25 threshold for "reasonable structure") suggest that the conflict/control distinction is a subtle direction in high-dimensional space rather than a gross geometric partition.

Layer-wise silhouette profiles. Full layer-wise silhouette scores for both models are reported in Table 10.

Table 10: Layer-wise cosine silhouette scores (PCA to 100 dimensions).

Layer	Llama 8B-Inst.	R1-Distill-Llama-8B
0	**[DATA]**	**[DATA]**
1	**[DATA]**	**[DATA]**
2	**[DATA]**	**[DATA]**
3	**[DATA]**	**[DATA]**
4	**[DATA]**	**[DATA]**
5	**[DATA]**	**[DATA]**
6	**[DATA]**	**[DATA]**
7	**[DATA]**	**[DATA]**
8	**[DATA]**	**[DATA]**
9	**[DATA]**	**[DATA]**
10	**[DATA]**	**[DATA]**
11	**[DATA]**	**[DATA]**
12	**[DATA]**	**[DATA]**
13	**[DATA]**	**[DATA]**
14	**[DATA]**	**[DATA]**
15	**[DATA]**	**[DATA]**
16	**[DATA]**	**[DATA]**
17	**[DATA]**	**[DATA]**
18	**[DATA]**	**[DATA]**
19	**[DATA]**	**[DATA]**
20	**[DATA]**	**[DATA]**
21	**[DATA]**	**[DATA]**
22	**[DATA]**	**[DATA]**
23	**[DATA]**	**[DATA]**
24	**[DATA]**	**[DATA]**
25	**[DATA]**	**[DATA]**
26	**[DATA]**	**[DATA]**
27	**[DATA]**	**[DATA]**
28	**[DATA]**	**[DATA]**
29	**[DATA]**	**[DATA]**

Layer	Llama 8B-Inst.	R1-Distill-Llama-8B
30	**[DATA]**	**[DATA]**
31	**[DATA]**	**[DATA]**

F.2 Centered Kernel Alignment (CKA)

CKA [Kornblith et al., 2019] measures the similarity of representational geometry between the two Llama-family models at each layer. A CKA value of 1.0 indicates identical representational structure (up to isotropic scaling); 0.0 indicates no linear relationship.

Layer-wise CKA. Full layer-wise CKA values between the two models:

Table 12: Layer-wise CKA (linear kernel) between Llama-3.1-8B-Instruct and R1-Distill-Llama-8B.

Layer	CKA
0	**[DATA]**
1	**[DATA]**
2	**[DATA]**
3	**[DATA]**
4	**[DATA]**
5	**[DATA]**
6	**[DATA]**
7	**[DATA]**
8	**[DATA]**
9	**[DATA]**
10	**[DATA]**
11	**[DATA]**
12	**[DATA]**
13	**[DATA]**
14	**[DATA]**
15	**[DATA]**
16	**[DATA]**
17	**[DATA]**
18	**[DATA]**
19	**[DATA]**
20	**[DATA]**
21	**[DATA]**
22	**[DATA]**
23	**[DATA]**
24	**[DATA]**
25	**[DATA]**
26	**[DATA]**
27	**[DATA]**
28	**[DATA]**
29	**[DATA]**
30	**[DATA]**
31	**[DATA]**

The CKA range (0.379–0.985) indicates that reasoning distillation preserves substantial representational structure at some layers while substantially reorganizing others. We expect that layers where CKA is lowest

(furthest from 1.0) correspond to where the deliberation gap in probe AUC is largest, as those are the layers most affected by reasoning training. Full layer-wise CKA values will test this prediction.

G. Lure Susceptibility Distribution

Lure susceptibility scores are continuous `P0` logit scores for the S1-lure answer on each conflict item. Positive scores indicate that the model's pre-generation representation favors the heuristic (lure) answer; negative scores indicate it favors the correct (normative) answer.

G.1 Distribution Summary

The sign flip between models is the key finding: Llama's pre-generation representation actively favors the lure (+0.422), while R1-Distill's favors the correct answer (−0.326). The effect size is small (Cohen's $d = 0.292$), reflecting the high variance within each model, but is statistically significant by both parametric and non-parametric tests.

G.2 Per-Category Breakdown

We expect the per-category breakdown to show that the sign flip is driven primarily by the vulnerable categories (base rate, conjunction, syllogism), where Llama shows high lure rates and large positive susceptibility scores. For the immune categories (CRT, arithmetic, framing, anchoring), both models should show negative susceptibility scores (favoring the correct answer). The per-category data will test whether the overall effect is driven by a few categories or represents a broad shift.

G.3 Histogram Description

The lure susceptibility distributions for both models are approximately normal with heavy tails. Llama's distribution is centered at +0.422 with standard deviation 2.986, indicating that the typical conflict item produces a moderate pro-lure bias in the pre-generation representation. R1-Distill's distribution is centered at −0.326 with standard deviation 2.056, shifted toward anti-lure bias with notably lower variance. The variance reduction in R1-Distill ($\sigma^2 = 4.23$ vs. 8.92 for Llama, F-test p **[DATA]**) suggests that reasoning distillation not only shifts the mean but also reduces the variability of lure susceptibility across items.

Placeholder key. All instances of **[DATA]** or **[D]** indicate values that will be filled when the full experimental results are available from GPU runs. The structural framework and all confirmed results are final.

References

Wim De Neys. Bias and conflict: A case for logical intuitions. *Perspectives on Psychological Science*, 7(1):28–38, 2012.

Gerd Gigerenzer and Ulrich Hoffrage. How to improve Bayesian reasoning without instruction: Frequency formats. *Psychological Review*, 102(4):684–704, 1995.

Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *Proceedings of the 36th International Conference on Machine Learning*, pages 3519–3529. PMLR, 2019.

Table 1: Full benchmark description. All items use novel surface features; classic versions are contamination baselines only. "Freq." = natural frequency framing subset.

Category	**# Pairs**	**Heuristic Family**	**Example Prompt (abbrev.)**	**Source**
Base rate neglect	35	Representativeness	"In a group of 995 engineers and 5 lawyers, a person is described as analytical... Is this person more likely a lawyer or engineer?"	Kahneman & Tversky (1973); novel isomorphs
Freq. subset	(10)	Representativeness	"10 out of 1000 people have condition X. A test has 95% sensitivity..."	Gigerenzer & Hoffrage (1995)
Conjunction fallacy	20	Representativeness	"Mx. Rivera is 31, outspoken, studied philosophy... Which is more probable: (A) bank teller, (B) bank teller and activist?"	Tversky & Kahneman (1983); novel isomorphs
Belief-bias syllogisms	25	Belief bias	"All flowers need water. Roses need water. Therefore, roses are flowers." (valid/invalid × believable/unbelievable)	Evans et al. (1983); novel isomorphs
Sunk cost fallacy	15	Loss aversion	"You have invested $9M in Project Aurora. A competitor has launched a superior product... Continue investing or cut losses?"	Arkes & Blumer (1985); novel isomorphs
CRT variants	30	Cognitive reflection	"A mug and a lid cost $1.10 in total. The mug costs $1.00 more than the lid..."	Frederick (2005); structural isomorphs
Arithmetic	25	Computation	"Calculate 247×18 step by step." (with/without carrying traps)	Original
Framing effects	20	Loss aversion	"Program A saves 200 people. Program B: 1/3 chance all 600 saved, 2/3 chance none saved." (gain/loss framing)	Tversky & Kahneman (1981); novel scenarios
Anchoring	20	Anchoring & adjustment	"The tallest building in the world is 828m. Estimate the average height of buildings in downtown Portland." (extreme vs. neutral anchor)	Tversky & Kahneman (1974); novel anchors
Classic baselines	**[N]**	Mixed	Original bat-and-ball, Linda, etc.	Contamination check only

Table 2: S1-lure rates (%) on conflict items by model and category, with binomial exact 95% confidence intervals in brackets. Bold indicates vulnerability (lure rate >10%). "—" indicates category not evaluated for that model.

	Llama	**R1-Distill**	**Qwen 3-8B**	
Category (n conflict)	**8B-Inst.**	**Llama-8B**	**no think**	**think**
Vulnerable categories				
Base rate (n=35)	**84** [68, 94]	4 [0, 13]	**56** [38, 72]	4 [0, 13]
Conjunction (n=20)	**55** [32, 77]	0 [0, 17]	**95** [75, 100]	**55** [32, 77]
Syllogism (n=25)	**52** [31, 72]	0 [0, 14]	0 [0, 14]	0 [0, 14]
Immune categories				
CRT variants (n=30)	0 [0, 12]	0 [0, 12]	0 [0, 12]	0 [0, 12]
Arithmetic (n=25)	0 [0, 14]	0 [0, 14]	0 [0, 14]	0 [0, 14]
Framing (n=20)	0 [0, 17]	0 [0, 17]	0 [0, 17]	0 [0, 17]
Anchoring (n=20)	0 [0, 17]	0 [0, 17]	0 [0, 17]	0 [0, 17]
Extended categories				
Sunk cost (n=15)	**[DATA]**	**[DATA]**	**[DATA]**	**[DATA]**
Overall (7-cat)	**27.3**	**2.4**	**21.0**	**7.0**

Table 3: Qwen 3-8B THINK vs. NO_THINK: per-category lure rates, accuracy, and response classification. The thinking toggle operates on identical weights via the `/think` and `/no_think` system prompt flags.

	NO_THINK			**THINK**		
Category	**Lure%**	**Correct%**	**Other%**	**Lure%**	**Correct%**	**Other%**
Base rate	56	**[DATA]**	**[DATA]**	4	**[DATA]**	**[DATA]**
Conjunction	95	**[DATA]**	**[DATA]**	55	**[DATA]**	**[DATA]**
Syllogism	0	**[DATA]**	**[DATA]**	0	**[DATA]**	**[DATA]**
CRT variants	0	**[DATA]**	**[DATA]**	0	**[DATA]**	**[DATA]**
Arithmetic	0	**[DATA]**	**[DATA]**	0	**[DATA]**	**[DATA]**
Framing	0	**[DATA]**	**[DATA]**	0	**[DATA]**	**[DATA]**
Anchoring	0	**[DATA]**	**[DATA]**	0	**[DATA]**	**[DATA]**
Overall	**21.0**	**[DATA]**	**[DATA]**	**7.0**	**[DATA]**	**[DATA]**

Table 5: Phase-wise summary of probe AUC. Phases defined as: early (L0–7), mid (L8–15), late (L16–23), final (L24–31). Gap = Llama − R1. Cohen's d and permutation p-value (10,000 permutations, BH-FDR corrected) reported for each phase.

Phase	**Layers**	**Llama** $\bar{\text{AUC}}$	**R1** $\bar{\text{AUC}}$	**Gap**	d	p	**Sig.**
Early	0–7	0.921	0.886	0.035	0.78	0.070	—
Mid	8–15	0.978	0.925	0.053	3.76	<0.001	*
Late	16–23	0.980	0.921	0.059	6.21	<0.001	*
Final	24–31	0.969	0.917	0.051	6.15	<0.001	*

Table 7: Cross-prediction AUC at tested layers. "Within" = probe AUC on vulnerable categories (same-category, different-fold). "Transfer" = probe AUC on immune categories. "Drop" = Within − Transfer.

	Llama 8B-Inst.			R1-Distill-Llama-8B		
Layer	**Within**	**Transfer**	**Drop**	**Within**	**Transfer**	**Drop**
0	0.848	0.702	0.146	0.782	0.734	0.048
4	0.927	0.439	0.488	0.919	0.878	0.041
8	0.952	0.384	0.568	0.920	0.878	0.042
12	0.988	0.449	0.539	0.927	0.728	0.199
14	**0.999**	**0.378**	**0.621**	**0.929**	**0.685**	**0.244**
16	0.995	0.569	0.426	0.929	0.696	0.233
20	0.978	0.386	0.592	0.914	0.615	0.299
24	0.978	0.326	0.652	0.924	0.524	0.400
28	0.981	0.365	0.616	0.923	0.415	0.508
31	0.954	0.438	0.516	0.904	0.385	0.519
Mean	0.960	0.444	0.516	0.907	0.654	0.253

Table 8: Category transfer matrix (Llama-3.1-8B-Instruct, L14). Rows = training category; columns = test category. Diagonal entries are within-category 5-fold CV AUC. Off-diagonal entries are transfer AUC (probe trained on row category, tested on column category). "Vuln." = vulnerable categories; "Imm." = immune categories.

	Vuln. (test)			Imm. (test)			
Train	**Base**	**Conj.**	**Syll.**	**CRT**	**Arith.**	**Fram.**	**Anch.**
Base rate	1.000	0.993	**[DATA]**	**[DATA]**	**[DATA]**	**[DATA]**	**[DATA]**
Conjunction	0.993	1.000	**[DATA]**	**[DATA]**	**[DATA]**	**[DATA]**	**[DATA]**
Syllogism	**[DATA]**	**[DATA]**	1.000	**[DATA]**	**[DATA]**	**[DATA]**	**[DATA]**
CRT	**[DATA]**	**[DATA]**	**[DATA]**	**[DATA]**	**[DATA]**	**[DATA]**	**[DATA]**
Arithmetic	**[DATA]**	**[DATA]**	**[DATA]**	**[DATA]**	**[DATA]**	**[DATA]**	**[DATA]**
Framing	**[DATA]**	**[DATA]**	**[DATA]**	**[DATA]**	**[DATA]**	**[DATA]**	**[DATA]**
Anchoring	**[DATA]**	**[DATA]**	**[DATA]**	**[DATA]**	**[DATA]**	**[DATA]**	**[DATA]**

Table 9: Cosine silhouette scores by layer phase. Positive values indicate meaningful clustering of conflict vs. no-conflict items. Values near zero indicate overlapping clusters. All values computed after PCA to 100 dimensions.

Model	**Peak Sil.**	**Peak Layer**	**Mean Sil. (L0–31)**	**Min Sil.**	**Min Layer**
Llama 8B-Inst.	0.079	**[DATA]**	**[DATA]**	**[DATA]**	**[DATA]**
R1-Distill	0.059	**[DATA]**	**[DATA]**	**[DATA]**	**[DATA]**

Table 11: CKA (linear kernel) between Llama-3.1-8B-Instruct and R1-Distill-Llama-8B residual stream activations at matched layers.

Summary	
Min CKA	0.379
Max CKA	0.985
Mean CKA	**[DATA]**

Table 13: Lure susceptibility score distributions. Scores are `P0` logit differences (lure − correct) on conflict items ($n = 142$ per model).

Model	**Mean**	**SD**	**Median**	**Min**	**Max**	n **positive**
Llama 8B-Inst.	+0.422	2.986	**[DATA]**	**[DATA]**	**[DATA]**	**[DATA]**
R1-Distill	−0.326	2.056	**[DATA]**	**[DATA]**	**[DATA]**	**[DATA]**

Welch's $t(250.2) = 2.459$, $p = 0.007$ (one-sided), Cohen's $d = 0.292$
Permutation test: $p = 0.008$ (10,000 permutations)

Table 14: Lure susceptibility scores by category and model. Positive = model favors lure; negative = model favors correct answer.

	Llama 8B-Inst.		**R1-Distill**	
Category	**Mean**	**SD**	**Mean**	**SD**
Base rate (n=35)	**[DATA]**	**[DATA]**	**[DATA]**	**[DATA]**
Conjunction (n=20)	**[DATA]**	**[DATA]**	**[DATA]**	**[DATA]**
Syllogism (n=25)	**[DATA]**	**[DATA]**	**[DATA]**	**[DATA]**
CRT (n=30)	**[DATA]**	**[DATA]**	**[DATA]**	**[DATA]**
Arithmetic (n=25)	**[DATA]**	**[DATA]**	**[DATA]**	**[DATA]**
Framing (n=20)	**[DATA]**	**[DATA]**	**[DATA]**	**[DATA]**
Anchoring (n=20)	**[DATA]**	**[DATA]**	**[DATA]**	**[DATA]**
Overall (n=142)	+0.422	2.986	−0.326	2.056